May 7

Cobb-Douglas matching function

$$m(U,V) = \omega \cdot U^{\eta} \cdot V^{1-\eta}$$

<u>job-finding rate:</u> $f(\theta) = \omega \cdot \theta^{1-\eta}$

<u>vacancy-filling rate:</u> $q(\theta) = \omega \cdot \theta^{-\eta}$

E: search effort (exogenous parameter)

Now, assume $E \neq 1$

$$m(U,V) = \omega \cdot (EU)^{\eta} \cdot V^{1-\eta}$$

$$\theta = \frac{V}{E \cdot U}$$ } aggreg. search effort

$f(\theta)$ and $q(\theta)$ do not change if we use the updated definition of tightness (θ)

$L^d(\theta) \rightarrow$ same as before

$$L^s(\theta) = \frac{E \cdot f(\theta)}{s + E f(\theta)} \cdot H$$

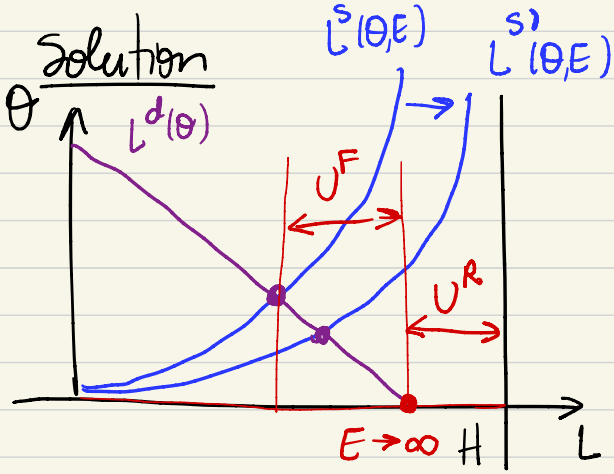
$\uparrow E \rightarrow \uparrow L^s$

(makes it easier for firms to hire)

$f(\theta) = \frac{m}{E \cdot U}$: rate at which workers find a job per unit of effort

rate of job-finding for effort E: $E \cdot f(\theta)$

Solution



$\uparrow E$: L^s moves to $L^{s'}$

$E \to \infty$

Alternatively, if we change the notation after the inclusion of E:

$$q(\theta) = w \cdot \theta^{-\eta} \cdot E^{\eta}$$

a change in E would impact both L^s and L^d, and it would be harder to identify which resulting change dominates the other.